

Unit 3 :: Complex Analysis

☞ PHY0400404: Mathematical Physics ☞

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Syllabus

Unit 3: Complex analysis

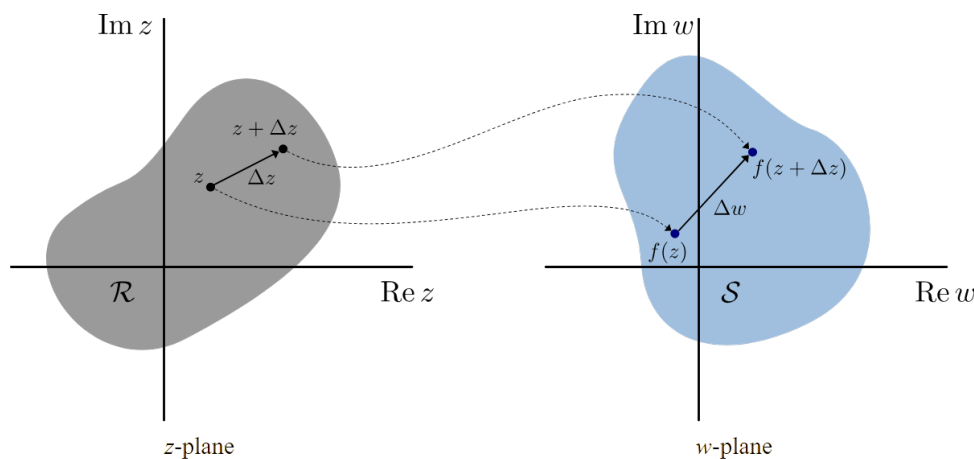
Functions of complex variables, analyticity and Cauchy-Riemann conditions, examples of analytic functions; singular functions: poles and branch points, order of singularity; integration of functions with complex variable, Cauchy's integral theorem and Cauchy's integral formula, simply and multiply connected regions; Laurent and Taylor's series expansions, residue theorem with application.

1 Functions of complex variables

Suppose, to each value that a complex variable z can assume, there corresponds one or more values of a complex variable w . We then say that w is a function of z and write $w = f(z)$.

- The complex variable z is called an *independent variable*
- The complex variable w is called a *dependent variable*.

1.1 Complex derivatives



If a complex function $f(z)$ is single-valued in some region $\mathcal{R}$ of the z plane, the derivative of $f(z)$ is defined as

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

provided that the limit exists independent of the manner in which Δz approaches 0. In such a case, we say that $f(z)$ is *differentiable* at z and $f'(z)$ is known as the *derivative* of the function $f(z)$.

1.2 Complex analytic functions

If the derivative $f'(z)$ exists at all points z of a region $\mathcal{R}$, then $f(z)$ is said to be analytic in $\mathcal{R}$ and is referred to as an *analytic function* in $\mathcal{R}$.

A function $f(z)$ is said to be analytic at a point z_0 if there exists a neighborhood $|z - z_0| < \delta$ at all points of which $f'(z)$ exists.

Note: Analytic functions are sometimes also known as *regular* or *holomorphic* functions.

1.3 Cauchy-Riemann conditions

Given a complex function which can be written as follows:

$$w = f(z) = u(x, y) + iv(x, y)$$

and is analytic over a region $\mathcal{R}$ on the z -plane, then the 2D functions $u(x, y)$ and $v(x, y)$ satisfy the following Cauchy-Riemann conditions in $\mathcal{R}$:

$$\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}}$$

If the partial derivatives of u and v are continuous in $\mathcal{R}$, then the Cauchy-Riemann equations are sufficient conditions that $f(z)$ be analytic in $\mathcal{R}$.

Note: The functions $u(x, y)$ and $v(x, y)$ are known as *conjugate functions*.

1.4 Examples of analytic functions

2 Singularities of complex functions

2.1 Poles

2.2 Branch points

3 Complex integration

3.1 Simply and multiply connected regions

3.2 Cauchy's integral theorem

3.3 Cauchy's integral formula

4 Complex series

4.1 Taylor's series expansion of a complex function

4.2 Laurent's series expansion of a complex function

4.2.1 Residue theorem